Claude API Connection Verification

check_setup.py -- Anthropic Python SDK setup and verification script

1. Objective

Verify that a working, correctly authenticated connection to the Claude API can be established using the official Anthropic Python SDK, with credentials loaded from environment configuration rather than hardcoded in source, and with clear, human-readable error handling for the most common setup mistakes.

2. What Was Built

check_setup.py loads ANTHROPIC_API_KEY and ANTHROPIC_MODEL from a local .env file using python-dotenv, sends a single short test message to Claude asking it to confirm the connection in one sentence, and prints three things on success:

- The model that actually answered (read from the API response, not just the request)
- Claude's reply text
- The input and output token counts from the response's usage data

If ANTHROPIC_API_KEY is missing, empty, or still set to an obvious placeholder value, the script prints a specific plain-English fix and exits -- it never attempts an API call with invalid credentials. If Anthropic's servers reject the key at request time (authentication failure), the script catches that specific error and prints a plain-English fix instead of letting a raw stack trace surface. Permission, model-not-found, and network-connection errors are handled the same way, each with its own targeted message.

3. Environment Setup Performed

- Installed Python 3.12 via winget (Python.Python.3.12)
- Installed the anthropic and python-dotenv packages via pip
- Created a local .env file (excluded from version control via .gitignore) to hold ANTHROPIC_API_KEY and ANTHROPIC_MODEL

4. Security Notes

- The API key is never hardcoded in check_setup.py -- it is read only from the environment at runtime
- The .env file is excluded from git via the project's .gitignore, so it cannot be committed accidentally
- No real API key value is included anywhere in this document or in the repository

5. How to Run

```
python check_setup.py
```

6. Full Script Listing -- check_setup.py

```
"""
check_setup.py - verifies the Claude API connection is configured correctly.

Loads ANTHROPIC_API_KEY and ANTHROPIC_MODEL from a .env file in this folder,
sends one short message to Claude, and prints the model that answered, the
reply, and the input/output token counts. Fails with a plain-English message
(never a raw stack trace) if the key is missing, empty, still a placeholder,
or rejected by Anthropic's servers.
"""

import os
import sys
from pathlib import Path

from dotenv import load_dotenv
import anthropic

ENV_PATH = Path(__file__).resolve().parent / ".env"

PLACEHOLDER_MARKERS = {
    "your-api-key-here",
    "your_api_key_here",
    "sk-ant-your-key-here",
    "replace_me",
    "replace-me",
    "changeme",
    "change-me",
    "xxxxxxxx",
    "placeholder",
    "example",
    "todo",
}


def looks_like_placeholder(value: str) -> bool:
    lowered = value.strip().lower()
    if lowered in PLACEHOLDER_MARKERS:
        return True
    if "your" in lowered and "key" in lowered:
        return True
    return False


def fail(message: str) -> None:
    print(message)
    sys.exit(1)


def main() -> None:
    load_dotenv(dotenv_path=ENV_PATH)

    api_key = os.environ.get("ANTHROPIC_API_KEY", "")
    model = os.environ.get("ANTHROPIC_MODEL", "claude-opus-5")
```

```
    if not api_key or not api_key.strip():
        fail(
            "No ANTHROPIC_API_KEY found.\n"
            f"Fix: create a file at {ENV_PATH} containing:\n"
            "  ANTHROPIC_API_KEY=sk-ant-...\n"
            "  ANTHROPIC_MODEL=claude-opus-5\n"
            "Get a real key from https://console.anthropic.com under API Keys."
        )

    if looks_like_placeholder(api_key):
        fail(
            "ANTHROPIC_API_KEY is still set to a placeholder value, not a real key.\n"
            f"Fix: open {ENV_PATH} and replace it with your real key from "
            "https://console.anthropic.com (API Keys section) - it starts with 'sk-ant-'."
        )

    client = anthropic.Anthropic(api_key=api_key)

    try:
        response = client.messages.create(
            model=model,
            max_tokens=100,
            messages=[{
                "role": "user",
                "content": "Reply with a single sentence confirming this connection works.",
            }],
        )
    except anthropic.AuthenticationError:
        fail(
            "Authentication failed - the API key was rejected by Anthropic's servers.\n"
            "Fix: the key in your .env file is invalid, revoked, or was mistyped when "
            "copied. Generate a fresh key at https://console.anthropic.com (API Keys) "
            f"and update {ENV_PATH}."
        )
    except anthropic.PermissionDeniedError:
        fail(
            "The API key was recognized but does not have permission for this request.\n"
            "Fix: check your organization's access/billing settings in the Anthropic Console."
        )
    except anthropic.NotFoundError:
        fail(
            f"Model '{model}' was not recognized.\n"
            f"Fix: check the ANTHROPIC_MODEL value in {ENV_PATH} - try 'claude-opus-5'."
        )
    except anthropic.APIConnectionError:
        fail(
            "Could not reach Anthropic's servers - this looks like a network problem.\n"
            "Fix: check your internet connection and try again."
        )

    reply_text = next((b.text for b in response.content if b.type == "text"), "")

    print(f"Model that answered: {response.model}")
    print(f"Reply: {reply_text}")
    print(f"Input tokens: {response.usage.input_tokens}")
    print(f"Output tokens: {response.usage.output_tokens}")
```

```
if __name__ == "__main__":
    main()
```